

Computer Fundamamental Assignments 1-5

Ques 01-05 2 Marks

Ques 06-10 4 Marks

Ques 11-15 8 Marks

Assignment 1

1. Name any two supercomputers of India. Also write where those supercomputers are located?
2. What do you mean by RAM? What type of memory it is?
3. What is sequential access memory? Write the names of any two memory elements which are based on sequential access concept.
4. What are the functions of ALU?
5. Define the term controller. Describe the function of keyboard controller.
6. Define the term Computer. Discuss various characteristics of computers.
7. What is concept keyboard? What are the advantages and disadvantages of using concept keyboards?
8. Differentiate between sequential access and direct access memory techniques. Give the examples of memory which are based on each of these principles.

9. Identify at least four attributes about printers and compare laser printers and inkjet printers on those attributes.
10. What is ROM? Give the classification of ROM and briefly write about each type.
11. Explain the architecture of computer system along with its components.
12. Classify various types of memory and explain each type of memory from the hierarchy.
13. Discuss the evolution of computers including the technology used in each generation of computers and the characteristics of the computers of those generations.
14. Describe various types of touch screen technologies and explain the working principle of each of them.
15. Explain the working principle of laser printers. Discuss how does a laser printer achieves high print quality compared to other printers.

Assignment 2

1. Explain system software, and how does it differ from application software?
2. Illustrate firmware and provide examples of devices that use firmware.
3. Compare the key differences between free software and proprietary software?
4. Show examples of popular free software applications
5. Show how does FOSS stand for, and why is it significant in the software

world?

6. Explain the need for an operating system in modern computing
7. Compare between batch processing, multiprogramming, and multitasking in the context of operating systems.
8. Show the need for an operating system in modern computing.
9. Summurize the essential services provided by an operating system.
10. Illustrate booting process of a computer with highlighting the roles of POST and BIOS.
11. Illustrate case study comparing MS-DOS, Windows, and a popular Linux distribution in terms of their features and usage.
12. Outline the role of operating systems evolved over time, and what are the major trends in modern operating systems?
13. Expalin the key components of a graphical user interface (GUI) in an operating system, and how do they enhance user interaction?
14. Show the security considerations and measures involved in the development and operation of operation of open-source software compared to proprietary software.
15. Illustrate the role of device drivers in an operating system, and why are they essential for hardware compatibility and functionality.

Assignment 3

1. Outline the key characteristics that differentiate Local Area Network (LANs). Metropolitan Area Networks (MANs), and Wide Area

Networks (WANs)? How do they serve different connectivity needs?

2. Describe the fundamental differences between the Internet and the World Wide Web (WWW).
3. What are the primary functions of networking elements such as hubs and switches in a network infrastructure?
4. In a network design scenario, discuss the factors that would lead you to choose either a hub or a switch for connecting devices
5. Explain What are the key characteristics and advantages of using twisted pair cables in network communication?
6. Compare and contrast coaxial cables and twisted pair cables as transmission media. What are the primary applications and advantages of coaxial cables in modern networking?
7. Explain the concept of bandwidth and signal attenuation in the context of different transmission media, including twisted pair, coaxial, and optical fiber cables. How do these factors affect the quality of data transmission?
8. What is a MAC (Media Access Control) address, and how does it relate to a network Interface card (NIC)?
9. Explain the significance of IP (Internet Protocol) addresses in computer networking.
10. In a typical client-server network setup, how do MAC and IP addresses work together to facilitate data transmission between devices on the same local Network and devices on different networks?
11. Describe the ISO-OS model and its seven layers. What are the primary functions of each layer, and how do they work together to

enable end-to-end communication in a network?

12. Explain the concept of network protocols in computer networking. How do protocols ensure effective communication between devices on a network, and what are some examples of common networking protocols?
13. Differentiate between viruses, worms, and malware. What are the key characteristics that set these malicious software types apart, and how do they propagate and damage systems differently?
14. Explore the various types of malware, including but not limited to viruses, worms, and Trojans. Provide examples of each type and discuss their specific purposes and risks to computer systems.
15. Explain what Spyware is and how it functions. What are the primary objectives of Spyware, and how does it compromise user privacy and system security?

Assignment 4

1. Explain Artificial intelligence and its application.
2. Explain Machine Learning and its application.
3. Explain emerging technologies and their applications: Cloud computing
4. Explain emerging technologies and their applications: Internet of Things
5. Explain emerging technologies and their applications: Blockchain

technology

6. Show How Artificial intelligence, Machine Learning, and Deep Learning differ from each other? ...
7. Explain types of Machine Learning?
8. Explain the use of computer vision in AI?
9. Explain knowledge representation in AI?
10. Show the different software platforms for AI development?
11. Explain Cloud Computing Services .
12. Explain software as a service, platform as a services?
13. Explain open source cloud computing platform databases?
14. Explain Internet of Things (IoT) in cloud computing?
15. Explain the difference between scalability and elasticity?

Assignment 5

1. What do you mean by RAM? What type of memory it is?
2. What are the functions of ALU?
3. Show examples of popular free software applications
4. Illustrate firmware and provide examples of devices that use firmware.
5. In a network design scenario, discuss the factors that would lead you

- to choose either a hub or a switch for connecting devices
6. Explain the significance of IP (Internet Protocol) addresses in computer networking.
 7. Compare and contrast coaxial cables and twisted pair cables as transmission media. What are the primary applications and advantages of coaxial cables in modern networking?
 8. Compare between batch processing, multiprogramming, and multitasking in the context of operating systems.
 9. Summurize the essential services provided by an operating system.
 10. Illustrate booting process of a computer with highlighting the roles of POST and BIOS
 11. Outline the role of operating systems evolved over time, and what are the major trends in modern operating systems?
 12. Describe various types of touch screen technologies and explain the working principle of each of them.
 13. Classify various types of memory and explain each type of memory from the hierarchy.
 14. Explain open source cloud computing platform databases? Explain Cloud Computing Services .
 15. Illustrate the role of device drivers in an operating system, and why are they essential for hardware compatibility and functionality.

Note - Underlined Questions are repeated Questions from previous assignment.

TPE Assignment 1-5

Ques 01-02 2 Marks

Ques 02-04 4 Marks

Ques 05 8 Marks

Assignment 1 & 2

1. State and explain active elements.
2. State and explain Kirchhoff.
3. Describe the dependent and independent sources in detail.
4. Explain the resistor, inductor and capacitor.
5. Describe the dependent and independent sources in detail.

Assignment 3

1. Convert the following number:
 - a. $(31)_{10}-(\dots\dots\dots)_2$
 - b. $(1111)_2-(\dots\dots\dots)_{10}$
2. State and prove De- Morgan's theorem.

3. Implement all the basic gates using universal gates.

4. Convert the given Boolean function into Minterm

$$Y = AB + BC + AC$$

5. Elaborate the basic theorems and laws of Boolean algebra and simplify the given expression:

$$Y = AB + AC + ABC(AB + C)$$

Assignment 4

1. Explain and classify the conductor, Insulator and Semiconductor.

a.) Describe semiconductor material with example. Explain its properties.

b) Describe Intrinsic and Extrinsic semiconductor.

2. Explain N Type and P Type Semiconductors.

3. Explain N Type and P Type Semiconductors.

4. Simplify the given Boolean expression using Boolean laws:

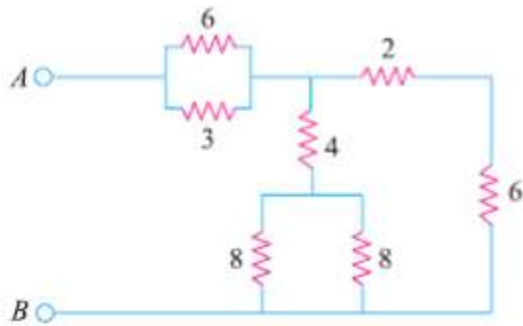
$$Y = ABC + ABC + ABC + ABC$$

5. Demonstrate the P-N junction with detail circuit diagram.

Assignment 5

1. Discuss the transformer losses.

2. Demonstrate series and parallel connection with suitable examples.
3. Describe the working principle and construction of a transformer in detail.
4. Describe the shell type and core type transformer.
5. Find the equivalent resistance of the circuit shown in figure.



Maths Assignment 2-5

Assignment 2

Question No.	Description	Marks
1	State the formula <u>radius of curvature</u> for parametric curves.	2
2	Find The Asymptotes of the following curves: $x^3 + yx^2 - xy^2 - y^3 - 3x - y - 1 = 0$.	2
3	Conclude the radius of curvature of the <u>curve</u> $y = a \sinh 2x$.	4
4	If $f(x) = 2x^3 - 9x^2 + 12x + 6$ Then Execute Maximum and Minimum Value in $[0,2]$	4
5	State and Prove the Leibnitz Theorem.	8

Assignment 4

Question No.	Description	Marks
1	Represent Beta Function.	2
2	Represent Gamma <u>Formula</u> .	2
3	Generate Riemann <u>Integration</u> of $\int_0^a 3x \, dx$.	4
4	Generate Riemann <u>Integration</u> of $\int_0^a x^2 \, dx$.	4
5	Evaluate $\iint x + y \, dx dy$ Where the region of integration is $3x + 2y \leq 1$ in the positive quadrant.	8

Assignment 5

Question No.	Description	Marks
1	State Stokes Theorem for Convergence.	2
2	Find the Curl of the Vector Field $\vec{P} = x^2i + yj + 2z k$ at the point $(1,2,2)$.	2
3	Find The Directional Derivative of $f(x, y, z) = x + yz^3$ in the Direction of the Vector $2i + 3j + 4k$ at the point $(0, -1, 1)$.	4
4	If V is the volume enclosed by a closed surface S and $\vec{F} = 2xi + yj + zk$ then Find $\int_V \vec{F} \cdot \hat{n} ds$.	4
5	If f is a homogeneous function of degree n , then Justify $x^2 \frac{\partial^3 f}{\partial x^3} + y^2 \frac{\partial^3 f}{\partial y^3} + 2xy \frac{\partial^3 f}{\partial x \partial y^2} = n(n-1)f.$	8

Assignment 3

Que-1 Explain partial differential with example

Ans A partial differential equation is a mathematical equation that involves partial derivative of a function with respect to the multiple independent variables.

For example, the heat equation, $\frac{\partial u}{\partial t} = \alpha \nabla^2 u$, describes how temperature u changes over time t in a substance.

Que-2 If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then prove $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

Ans

Given

$$u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} (3x^2 - 3yz)$$

$$\text{Here } \frac{d}{dx}(\log x) = \frac{1}{x}, \quad x > 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\text{and } \frac{d}{dx}(\text{constant}) = 0$$

$$\Rightarrow \frac{\partial v}{\partial x} = \frac{1}{\log(x^3 + y^3 + z^3 - 3xyz)} \cdot (3x^2 + 0 + 0 - 3yz)$$

$$\Rightarrow \frac{\partial v}{\partial x} = \frac{3x^2 - 3yz}{\log(x^3 + y^3 + z^3 - 3xyz)} \quad \text{--- (1)}$$

$\Rightarrow$ Similarly,

$$\Rightarrow \frac{\partial v}{\partial y} = \frac{3y^2 - 3xz}{\log(x^3 + y^3 + z^3 - 3xyz)} \quad \text{--- (2)}$$

$$\Rightarrow \frac{\partial v}{\partial z} = \frac{3z^2 - 3xy}{\log(x^3 + y^3 + z^3 - 3xyz)} \quad \text{--- (3)}$$

Now,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3x^2 - 3yz + 3y^2 - 3xz + 3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{3(x^2 + y^2 + z^2 - xy + yz + xz)}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{3(x^2 + y^2 + z^2 - xy + yz + xz)}{(x+y+z)(x^2 + y^2 + z^2 - xy + yz + xz)}$$

$$\therefore x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2+y^2+z^2 - xy - yz - zx)$$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$$

Hence proved

Que-3 If $y = \sin^{-1} \frac{x^2 + y^2}{x+y}$, then prove that

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \tan(u)$$

Solve

$$\sin u = \frac{x^2 + y^2}{x+y}$$

differentiating with respect to x

$$\text{we get, } \cos u \cdot \frac{du}{dx} = \frac{2x(xy) - (x^2+y)^2}{(x+y)^2}$$

$$\cos u \cdot \frac{du}{dx} = \frac{2y(xy) - x^2 + y^2}{(x+y)^2}$$

$$= \frac{x \cdot \frac{dy}{dx} + y \cdot \frac{du}{dy} - \sin u \left(\frac{y^3 + y^2 + x^2}{(x+y)^2} \right)}{}$$

$$= \sec u \cdot \sin u$$

$$= \tan(u)$$

Ques 4 If f is a homogeneous function of degree n , then.

$$y \frac{\partial^2 f}{\partial x^2} + x \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial y}$$

Solve let (By Euler's theorem)

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf \quad \text{--- (1)}$$

By eq (1) the relative partial differentiation of x , then,

$$\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{n \partial f}{\partial x}$$

$$\Rightarrow x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial x}$$

Again in eq (1) the relative partial differentiation of y , then

$$x \cdot \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial y^2} = \frac{n \partial f}{\partial y}$$

$$= x \cdot \frac{\partial^2 f}{\partial y \partial x} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial y}$$

Ques If f is a homogenous function of degree n , then

$$x^2 \frac{d^2 f}{dx^2} + y^2 \frac{d^2 f}{dy^2} + 2xy \frac{d^2 f}{dx dy} = n(n-1)f$$

Ans let by Euler's law

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f \quad \text{--- (1)}$$

By eq (1) the relative partial differentiation w.r.t x , then,

$$\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = n \frac{\partial f}{\partial x}$$

$$\Rightarrow x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial x}$$

Again the eq (1) the relative partial differentiation w.r.t y , then,

$$x \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial y^2} = n \frac{\partial f}{\partial y}$$

$$\Rightarrow x \frac{\partial^2 y}{\partial y \partial x} + y \frac{\partial^2 f}{\partial y^2} = (n-1) \frac{\partial f}{\partial y}$$

Now eq (2) multiply by x and eq (3) multiply by y and then sum of both we get,

$$x^2 \frac{\partial^2 f}{\partial x^2} + xy \frac{\partial^2 f}{\partial x \partial y} + xy \frac{\partial^2 f}{\partial y \partial x} + y^2 \frac{\partial^2 f}{\partial y^2}$$

$$= (n-1) x \frac{\partial f}{\partial x} + (n-1) y \frac{\partial f}{\partial y}$$

$$\Rightarrow x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} =$$

$$(n-1) \left[x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right]$$

$$\left(\because \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \right)$$

$$x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = n(n-1) f$$

from eq (1)

Hence proved Ans

